Supplementary Material: Adversarial Contingency Auctions for Decentralized Multi-Robot Task Allocation under Strategic Blockages

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I. Overview

This supplementary document provides extended statistical visualizations and diagnostics supporting the primary manuscript. All evaluations correspond to the pre-registered benchmark sweep ($N = 250$ matched independent instances, $13,800$ total simulation trials) conducted across 5 graph families, 4 adversary behavioral profiles, and 3 communication regimes.

II. Extended Performance Distributions

Fig. 1 illustrates the full sample distribution across all primary metrics via raincloud plots combining half-violin kernel density estimates, boxplots with interquartile ranges, and jittered instance points.

III. Non-Parametric Effect Size Heatmap

Fig. 2 details pairwise effect sizes evaluated via Cliff's δ non-parametric metric across all algorithm pairings.

IV. Confidence Interval Width Scaling

Fig. 3 validates statistical sample size adequacy by evaluating empirical 95% bootstrap confidence interval half-widths as a function of sample instance count $N \in [25, 250]$.

V. Rollout Execution Keyframes

Fig. 4 provides synchronized execution keyframes on the Downtown Boston road network comparing ACA against CTP Replan and standard CBBA under identical adversarial interdictions.

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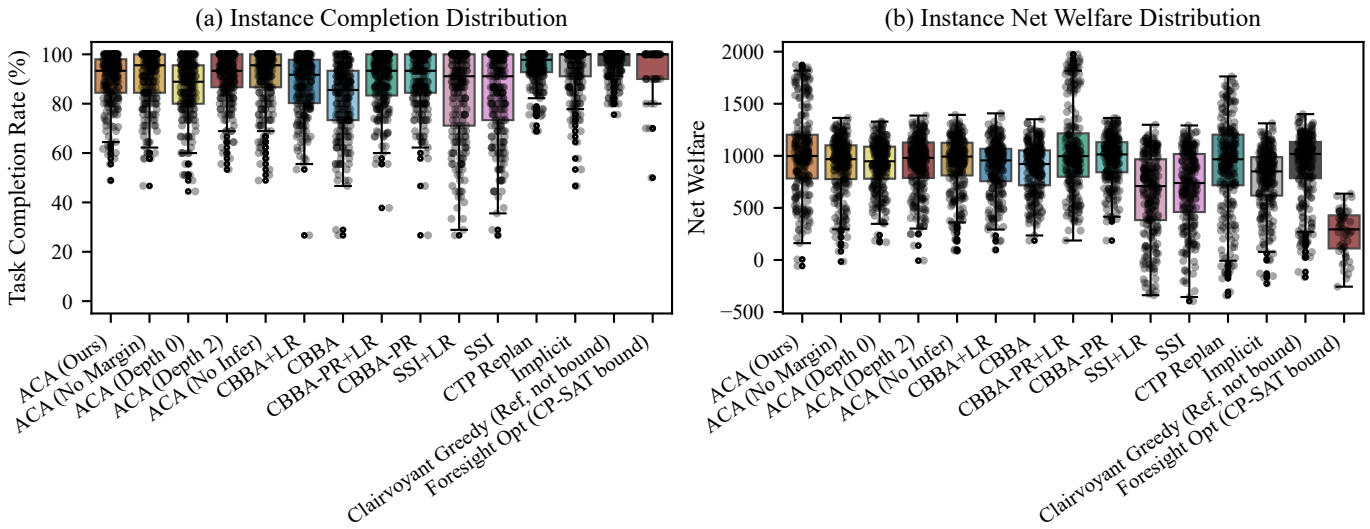


Fig. 1. Raincloud Performance Distributions ($N = 250$ matched instances per method across 5 graph topologies): Panels depict mission completion rate (%), net team welfare, distinct task conflict counts, and total communication message volume. Boxplots show median and interquartile ranges; half-violins illustrate kernel probability densities; points indicate individual instance evaluations.

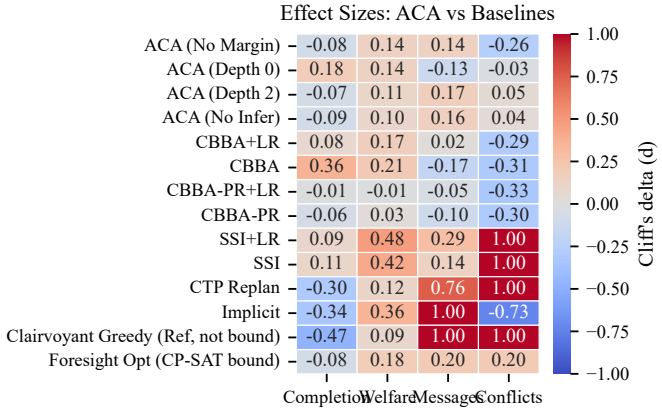


Fig. 2. Pairwise Non-Parametric Effect Size Heatmap (Cliff's δ, $N = 250$ matched instances per comparison): Color gradient indicates magnitude and direction of difference (warm: ACA advantage; cool: baseline advantage). Significance evaluated via Mann–Whitney U test with Holm–Bonferroni correction.

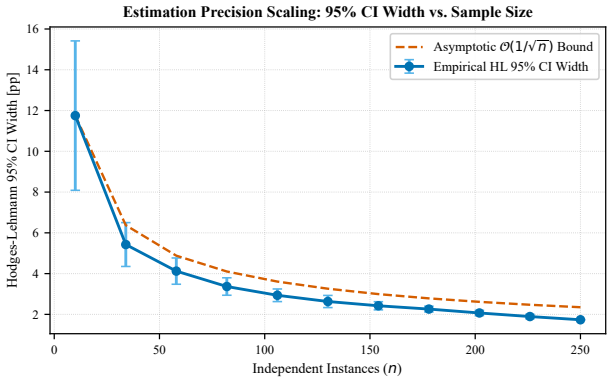


Fig. 3. Confidence Interval Width Scaling ($10,000$ bootstrap resamples across subsample sizes $N \in [25, 250]$): Shaded regions depict empirical 95% bootstrap CI bounds on mission completion rate, demonstrating asymptotic error reduction consistent with $O(1/\sqrt{N})$ central limit convergence.

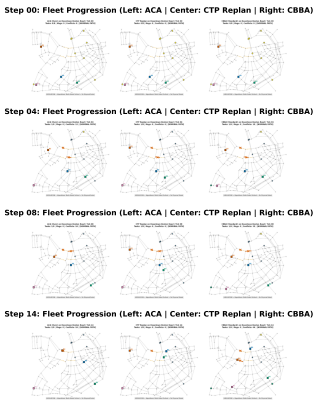


Fig. 4. Synchronized Rollout Execution Keyframes on Downtown Boston Road Network ($n = 216, |E| = 323$) under Identical Adversarial Blockage Stream: Columns depict ACA (Ours, Left), CTP Replan (Center), and Standard CBBA (Right) across time steps $t \in \{0, 4, 8, 14\}$. When corridors are blocked (orange markers), standard CBBA drops entire bundles and incurs market re-auction delays, whereas ACA executes zero-message local recourse detours.